Assessment Inequity and Land Value Taxation: A Multi-City Analysis of Property Tax Disparities in Cook County, New York City, and Philadelphia

Krishna Singh Isabelle Byrne

Northeastern University
Boston, United States

MATH 7339: Machine Learning and Statistical Learning Theory II
Instructor: He Wang

Spring 2026

Outline

1. Motivation & Research Questions
2. Data
3. Methodology
4. Results
5. Discussion & Conclusions

Motivation

- **The problem:** Property taxes are levied on *assessed* value, not true market value. Assessment errors translate directly into tax errors.

Motivation

- **The problem:** Property taxes are levied on *assessed* value, not true market value. Assessment errors translate directly into tax errors.
- **Known pattern:** Lower-value properties are routinely assessed at *higher fractions* of market value than higher-value properties, a phenomenon called **assessment regressivity**.

Motivation

- **The problem:** Property taxes are levied on *assessed* value, not true market value. Assessment errors translate directly into tax errors.
- **Known pattern:** Lower-value properties are routinely assessed at *higher fractions* of market value than higher-value properties, a phenomenon called **assessment regressivity**.
- **Racial dimension:** Majority-Black neighborhoods in Chicago, Detroit, and New Orleans carry effective tax burdens well above those of predominantly white neighborhoods at comparable values.

Motivation

- **The problem:** Property taxes are levied on *assessed* value, not true market value. Assessment errors translate directly into tax errors.
- **Known pattern:** Lower-value properties are routinely assessed at *higher fractions* of market value than higher-value properties, a phenomenon called **assessment regressivity**.
- **Racial dimension:** Majority-Black neighborhoods in Chicago, Detroit, and New Orleans carry effective tax burdens well above those of predominantly white neighborhoods at comparable values.
- **A proposed remedy:** A **land-value tax (LVT)** (taxing only the unimproved site value, not buildings) is predicted by theory to reduce speculative holding, incentivize construction, and lower housing costs.

Motivation

- **The problem:** Property taxes are levied on *assessed* value, not true market value. Assessment errors translate directly into tax errors.
- **Known pattern:** Lower-value properties are routinely assessed at *higher fractions* of market value than higher-value properties, a phenomenon called **assessment regressivity**.
- **Racial dimension:** Majority-Black neighborhoods in Chicago, Detroit, and New Orleans carry effective tax burdens well above those of predominantly white neighborhoods at comparable values.
- **A proposed remedy:** A **land-value tax (LVT)** (taxing only the unimproved site value, not buildings) is predicted by theory to reduce speculative holding, incentivize construction, and lower housing costs.
- **Key question:** Would a revenue-neutral LVT actually *reduce* these disparities, or does historical disinvestment cause it to fall hardest on majority-Black neighborhoods?

Research Questions

Three cities analyzed: Cook County (IL), New York City (NY), Philadelphia (PA)

1. **Assessment equity:** Do official assessed values systematically over-assess properties in lower-income and majority-Black neighborhoods relative to model-predicted market values?

Research Questions

Three cities analyzed: Cook County (IL), New York City (NY), Philadelphia (PA)

1. **Assessment equity:** Do official assessed values systematically over-assess properties in lower-income and majority-Black neighborhoods relative to model-predicted market values?
2. **LVT distributional effects:** Would a revenue-neutral land-value-only tax reduce disparities in effective tax rates across racial and income groups?

Research Questions

Three cities analyzed: Cook County (IL), New York City (NY), Philadelphia (PA)

1. **Assessment equity:** Do official assessed values systematically over-assess properties in lower-income and majority-Black neighborhoods relative to model-predicted market values?
2. **LVT distributional effects:** Would a revenue-neutral land-value-only tax reduce disparities in effective tax rates across racial and income groups?
3. **Neighborhood typology:** Which neighborhood types bear the burden of the current system, and which benefit most from the simulated reform?

Research Questions

Three cities analyzed: Cook County (IL), New York City (NY), Philadelphia (PA)

1. **Assessment equity:** Do official assessed values systematically over-assess properties in lower-income and majority-Black neighborhoods relative to model-predicted market values?
2. **LVT distributional effects:** Would a revenue-neutral land-value-only tax reduce disparities in effective tax rates across racial and income groups?
3. **Neighborhood typology:** Which neighborhood types bear the burden of the current system, and which benefit most from the simulated reform?

Motivation: Frederiksen hypothesis – if Cook County taxing bodies shifted to site-only taxation, the *majority* of homeowners would pay lower property taxes. We test this directly.

Data Sources

Cook County, IL

- CCAO open data portal + parcel universe
- 10 spatial features: CTA/Metra distance, flood risk, foreclosure density, noise
- Sales: $10K–$10M, 2021–2023
- **1,635,253** residential parcels
- **127,549** qualifying sales
- Median sale price: $325K

Philadelphia, PA

- OPA assessment database
- Sales: $10K–$10M, 2019+
- **504,034** residential parcels

New York City, NY

- NYC MapPLUTO (Dept. of City Planning)
- Finance rolling sales files
- Sales: $10K–$50M
- **705,173** residential parcels
- 565,301 Tax Class 1 (1–3 family)
- 139,872 Tax Class 2 (apartments, co-ops)
- **114,754** qualifying sales
- Median sale price: $785K

Demographics (all cities)

- Census ACS 5-year (2019–2023)
- Merged via census tract GEOID

Automated Valuation Models (AVM)

Goal: Predict market value for every residential parcel to serve as the ground truth for assessment ratio analysis.

Automated Valuation Models (AVM)

Goal: Predict market value for every residential parcel to serve as the ground truth for assessment ratio analysis.

- **Models trained:** OLS, Random Forest, XGBoost, LightGBM
- **Target:** log(sale price), winsorized at 1st/99th percentile; predictions exponentiated back to dollars
- **Split:** 80% train / 20% test; preprocessing fit on train only

Automated Valuation Models (AVM)

Goal: Predict market value for every residential parcel to serve as the ground truth for assessment ratio analysis.

- **Models trained:** OLS, Random Forest, XGBoost, LightGBM
- **Target:** log(sale price), winsorized at 1st/99th percentile; predictions exponentiated back to dollars
- **Split:** 80% train / 20% test; preprocessing fit on train only

Feature sets by city:

- **Cook County:** sq. ft., age, beds, baths, garage, construction quality + **10 spatial features** (transit distance, flood risk, foreclosure density, lake distance, noise contour)
- **NYC:** gross area, lot area, age, floors, residential units (Class 1 only; Class 2 excluded per RPTL §581)
- **Philadelphia:** livable area, total area, age, beds, baths, rooms

Automated Valuation Models (AVM)

Goal: Predict market value for every residential parcel to serve as the ground truth for assessment ratio analysis.

- **Models trained:** OLS, Random Forest, XGBoost, LightGBM
- **Target:** log(sale price), winsorized at 1st/99th percentile; predictions exponentiated back to dollars
- **Split:** 80% train / 20% test; preprocessing fit on train only

Feature sets by city:

- **Cook County:** sq. ft., age, beds, baths, garage, construction quality + **10 spatial features** (transit distance, flood risk, foreclosure density, lake distance, noise contour)
- **NYC:** gross area, lot area, age, floors, residential units (Class 1 only; Class 2 excluded per RPTL §581)
- **Philadelphia:** livable area, total area, age, beds, baths, rooms

Note: Cook County has 10 spatial features unavailable for NYC and Philadelphia, so cross-city AVM comparisons are partially confounded by feature availability.

IAAO Assessment Equity Metrics

Assessment ratio for parcel i:

$$r_i = \frac{\text{implied market value (from AV)}}{\text{predicted market value}}$$

Ratio > 1: over-assessed. Ratio < 1: under-assessed.

IAAO Assessment Equity Metrics

Assessment ratio for parcel i:

$$r_i = \frac{\text{implied market value (from AV)}}{\text{predicted market value}}$$

Ratio > 1: over-assessed. Ratio < 1: under-assessed.

Three IAAO ratio statistics [1]:

- **COD** (Coefficient of Dispersion): measures spread in ratios.

$$\mathrm{COD} = \frac{\overline{|r_i - \tilde{r}|}}{\tilde{r}} \times 100 \qquad \text{Standard: 5\%–15\%}$$

- **PRD** (Price-Related Differential): tests regressivity by value.

$$\mathrm{PRD} = \frac{\bar{r}}{\sum r_i s_i / \sum s_i} \qquad \text{Standard: 0.98–1.03}$$

- **PRB** (Price-Related Bias): regression-based regressivity test.

$$\frac{r_i - \tilde{r}}{\tilde{r}} = \alpha + \beta\,[0.5\ln(s_i) + 0.5\ln(AV_i)] + \varepsilon_i \qquad \text{Standard: } |\beta| < 0.05$$

Significant negative $\beta \Rightarrow$ regressivity.

LVT Simulation: Revenue-Neutral Design

Revenue neutrality: total tax revenue held constant by adjusting the rate applied to the land-only base.

LVT Simulation: Revenue-Neutral Design

Revenue neutrality: total tax revenue held constant by adjusting the rate applied to the land-only base.

Revenue-neutral multiplier:

$$m = \frac{\sum_i AV_i^{\text{total}} \times r_i}{\sum_i AV_i^{\text{land}} \times r_i}$$

LVT Simulation: Revenue-Neutral Design

Revenue neutrality: total tax revenue held constant by adjusting the rate applied to the land-only base.

Revenue-neutral multiplier:

$$m = \frac{\sum_i AV_i^{\text{total}} \times r_i}{\sum_i AV_i^{\text{land}} \times r_i}$$

Per-parcel bills (e = equalization factor):

$$T_i^{\text{current}} = AV_i^{\text{total}} \times e \times r_i$$
$$T_i^{\text{LVT}} = AV_i^{\text{land}} \times e \times r_i \times m$$

LVT Simulation: Revenue-Neutral Design

Revenue neutrality: total tax revenue held constant by adjusting the rate applied to the land-only base.

Revenue-neutral multiplier:

$$m = \frac{\sum_i AV_i^{\text{total}} \times r_i}{\sum_i AV_i^{\text{land}} \times r_i}$$

Per-parcel bills (e = equalization factor):

$$T_i^{\text{current}} = AV_i^{\text{total}} \times e \times r_i$$
$$T_i^{\text{LVT}} = AV_i^{\text{land}} \times e \times r_i \times m$$

- **Cook County:** PIN-level rates from Cook County Clerk (4,464 codes; 1.7%–32.1%; median 7.52%); $e = 3.0163$
- **NYC & Philadelphia:** city-wide uniform rates
- Revenue deviation: $<$**0.01%** all cities (floating-point only)
- **LVT beneficiary:** $T_i^{\text{LVT}} < T_i^{\text{current}}$

Classification & Clustering Methods

LVT Benefit Classification

Predict parcel-level binary outcome: does the parcel pay less under LVT?

- Feature set: total market value, assessment ratio, building size, age, beds, baths + tract demographics (income, % Black, % Hispanic, % owner-occupied)
- *Market value land/bldg excluded* – including them encodes the outcome directly
- Three classifiers: **Logistic Regression**, **Linear SVM** (CalibratedClassifierCV), **MLP** (128-64, ReLU, early stopping)
- Primary metric: **AUC** (benefit rates differ across cities)

Classification & Clustering Methods

LVT Benefit Classification

Predict parcel-level binary outcome: does the parcel pay less under LVT?

- Feature set: total market value, assessment ratio, building size, age, beds, baths + tract demographics (income, % Black, % Hispanic, % owner-occupied)
- *Market value land/bldg excluded* – including them encodes the outcome directly
- Three classifiers: **Logistic Regression**, **Linear SVM** (CalibratedClassifierCV), **MLP** (128-64, ReLU, early stopping)
- Primary metric: **AUC** (benefit rates differ across cities)

Neighborhood Clustering (GMM)

- Aggregate parcel outcomes to Census tracts
- Features: median ratio, land ratio, % tax change, % benefit, income, % Black, % Hispanic, % owner-occupied, median predicted value
- PCA preprocessing ($\geq$ 90% variance retained); BIC for component selection
- Gaussian Mixture Models – soft assignments, full covariance

AVM Performance

City	Model	R^2	RMSE	MAE	MdAPE
Cook County	OLS	0.656	\$172K	\$101K	19.5%
	RF	0.787	\$136K	\$83K	15.9%
	XGBoost	0.812	\$127K	\$79K	15.4%
	LightGBM	**0.817**	**\$126K**	**\$78K**	**15.2%**
NYC	OLS		*degenerate – excluded*		
	RF	0.557	\$347K	\$209K	15.6%
	LightGBM	**0.572**	**\$341K**	**\$207K**	**15.6%**
Philly	OLS		*degenerate – excluded*		
	RF	0.680	\$104K	\$70K	22.5%
	LightGBM	**0.735**	**\$94K**	**\$63K**	**20.4%**

AVM Performance

City	Model	R^2	RMSE	MAE	MdAPE
Cook County	OLS	0.656	\$172K	\$101K	19.5%
	RF	0.787	\$136K	\$83K	15.9%
	XGBoost	0.812	\$127K	\$79K	15.4%
	LightGBM	**0.817**	**\$126K**	**\$78K**	**15.2%**
NYC	OLS		*degenerate – excluded*		
	RF	0.557	\$347K	\$209K	15.6%
	LightGBM	**0.572**	**\$341K**	**\$207K**	**15.6%**
Philly	OLS		*degenerate – excluded*		
	RF	0.680	\$104K	\$70K	22.5%
	LightGBM	**0.735**	**\$94K**	**\$63K**	**20.4%**

- Cook County $R^2 = 0.817$: competitive with CCAO production model ($R^2 \approx 0.85$–0.90)
- NYC lower R^2 reflects heterogeneous market *and* missing spatial features vs. Cook County

IAAO Ratio Study Results

City	Med. Ratio	COD	PRD	PRB
Cook County	0.884	28.1%†	1.093†	+0.015
New York City	1.150	39.9%†	1.249†	−1.048†
Philadelphia	1.091	42.8%†	1.255†	−0.492†

† Outside IAAO acceptable range.

Standards: COD 5–15%, PRD 0.98–1.03, PRB ±0.05.

IAAO Ratio Study Results

City	Med. Ratio	COD	PRD	PRB
Cook County	0.884	28.1%†	1.093†	+0.015
New York City	1.150	39.9%†	1.249†	−1.048†
Philadelphia	1.091	42.8%†	1.255†	−0.492†

† Outside IAAO acceptable range.
Standards: COD 5–15%, PRD 0.98–1.03, PRB ±0.05.

- **All three cities fail** IAAO standards for COD and PRD
- NYC PRB $= -1.048$ ($p < 0.001$) and Philly PRB $= -0.492$ ($p < 0.001$): **statistically significant regressivity**, lower-value properties assessed at higher ratios
- **Cook County PRB** $= +0.015$: within acceptable range, reflecting the CCAO's equity-focused modeling approach

Assessment Equity by Racial Group

Group	Cook County	NYC (Class 1)	Philadelphia
Majority Black	0.630	1.048	1.213
Majority Hispanic	0.776	1.165	1.479
Mixed	0.764	1.230	1.205
Majority White	0.816	1.329	1.159

Assessment Equity by Racial Group

Group	Cook County	NYC (Class 1)	Philadelphia
Majority Black	0.630	1.048	1.213
Majority Hispanic	0.776	1.165	1.479
Mixed	0.764	1.230	1.205
Majority White	0.816	1.329	1.159

Cook County: 18.6 pp Black-White gap. After controlling for spatial features, the racial coefficient is *not* significant ($p = 0.382$), so disparities operate through *place-based disinvestment*, not direct assessor bias.

Assessment Equity by Racial Group

Group	Cook County	NYC (Class 1)	Philadelphia
Majority Black	0.630	1.048	1.213
Majority Hispanic	0.776	1.165	1.479
Mixed	0.764	1.230	1.205
Majority White	0.816	1.329	1.159

Cook County: 18.6 pp Black-White gap. After controlling for spatial features, the racial coefficient is *not* significant ($p = 0.382$), so disparities operate through *place-based disinvestment*, not direct assessor bias.
NYC: 28.1 pp Black-White gap. OLS confirms `pct_black` $= -0.0044$ ($p < 0.001$). Driven by the statutory *assessment cap mechanism*: rapidly appreciating minority neighborhoods are capped below the 6% target.

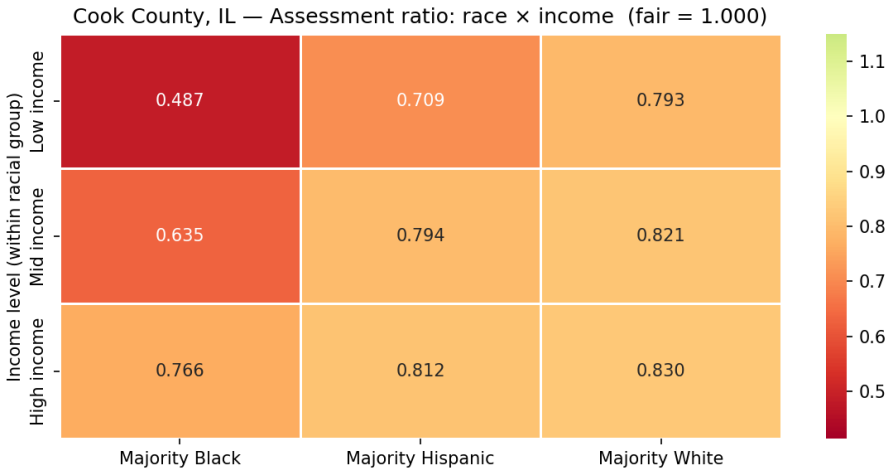
Assessment Equity by Racial Group

Group	Cook County	NYC (Class 1)	Philadelphia
Majority Black	0.630	1.048	1.213
Majority Hispanic	0.776	1.165	1.479
Mixed	0.764	1.230	1.205
Majority White	0.816	1.329	1.159

Cook County: 18.6 pp Black-White gap. After controlling for spatial features, the racial coefficient is *not* significant ($p = 0.382$), so disparities operate through *place-based disinvestment*, not direct assessor bias.
NYC: 28.1 pp Black-White gap. OLS confirms `pct_black` $= -0.0044$ ($p < 0.001$). Driven by the statutory *assessment cap mechanism*: rapidly appreciating minority neighborhoods are capped below the 6% target.
Philadelphia: All groups above 1.0 (systematic over-assessment). No significant racial coefficient ($p = 0.287$), but only 245 tracts; model is underpowered.

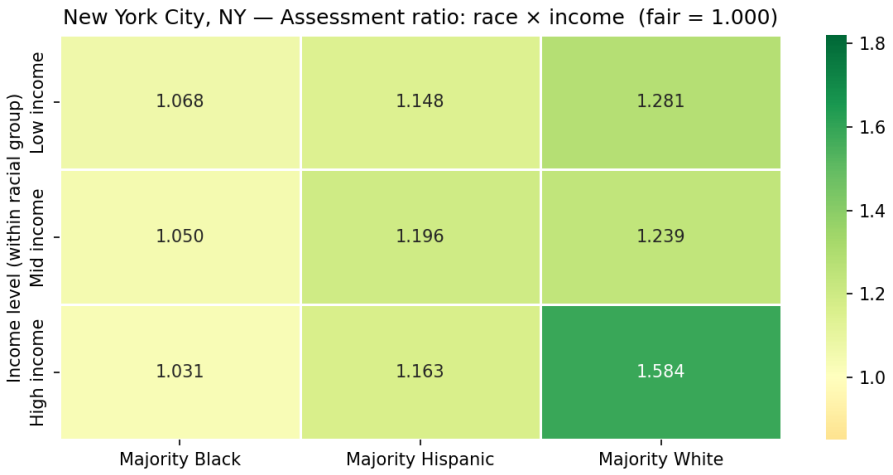
Intersectional Assessment Ratios: Race × Income

Cook County



Low-income Black: **0.487**
Low-income White: 0.793

New York City (Class 1)



Low-income Black: 1.068
High-income White: **1.584**
(opposite pattern – cap mechanism)

The racial gap persists within income brackets in Cook County. In NYC, the pattern reverses due to the statutory assessment cap: rapidly appreciating minority neighborhoods are capped below target.

LVT Simulation Results

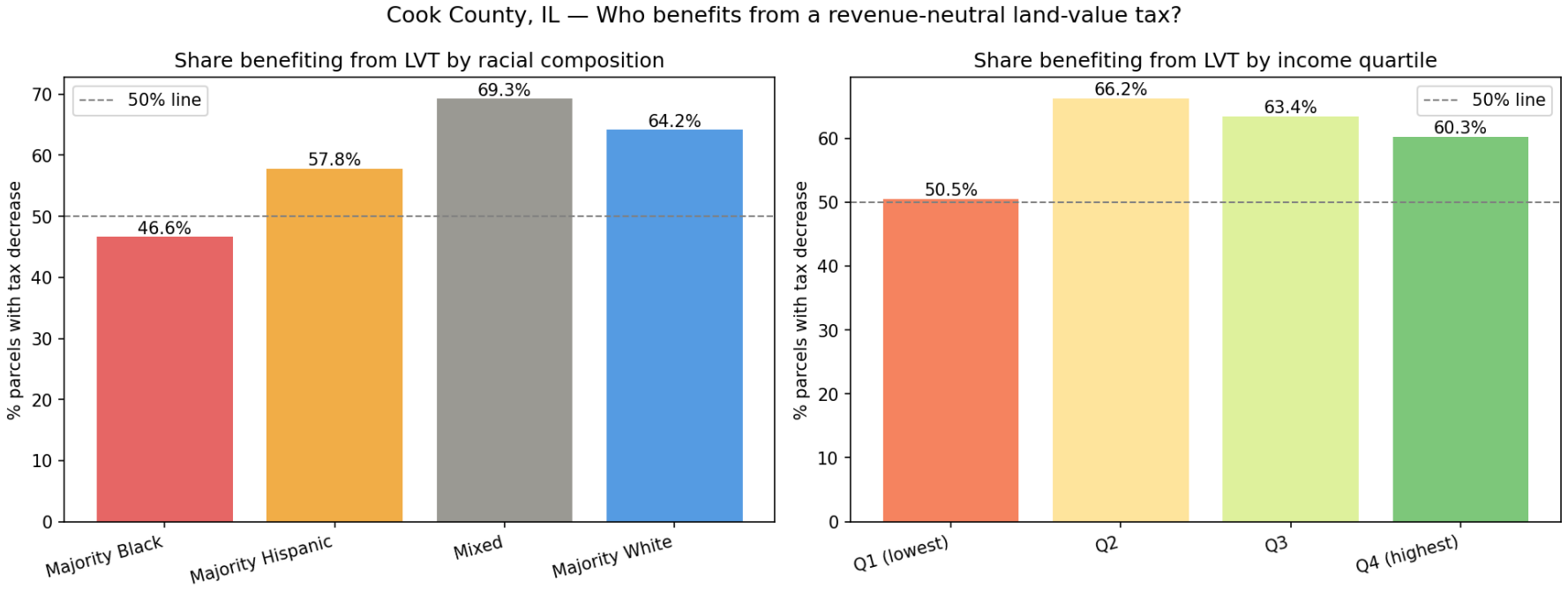
		Cook County	NYC	Philadelphia
LVT multiplier		4.40×	5.67×	3.55×
% benefit	Overall	**61.3%**	**30.3%**	**58.2%**
	Maj. Black	46.6%	18.8%	52.3%
	Maj. Hispanic	57.8%	38.6%	59.0%
	Mixed	69.3%	30.0%	59.2%
	Maj. White	64.2%	35.2%	64.2%
Gini current		0.396	0.444	0.279
Gini LVT		0.433	0.458	0.186

LVT Simulation Results

		Cook County	NYC	Philadelphia
LVT multiplier		4.40×	5.67×	3.55×
% benefit	Overall	**61.3%**	**30.3%**	**58.2%**
	Maj. Black	46.6%	18.8%	52.3%
	Maj. Hispanic	57.8%	38.6%	59.0%
	Mixed	69.3%	30.0%	59.2%
	Maj. White	64.2%	35.2%	64.2%
Gini current		0.396	0.444	0.279
Gini LVT		0.433	0.458	0.186

- **Frederiksen hypothesis holds** in Cook County (61.3%) and Philadelphia (58.2%) overall, but **fails for majority-Black neighborhoods in all three cities**
- Black-White benefit gaps: **17.6 pp** (Cook), **16.4 pp** (NYC), **11.9 pp** (Philly)
- NYC: LVT multiplier of 5.67× reflects extreme land scarcity; nearly all parcels see higher bills

Who Benefits from LVT?



Majority-Black neighborhoods are the **only group below the 50% line** in Cook County. Lower-income quartiles also benefit at lower rates, reflecting concentration of high land-to-improvement ratio parcels in disinvested areas.

Why Majority-Black Neighborhoods Lose

The structural mechanism:

Decades of redlining and disinvestment have left majority-Black neighborhoods with:

- Deteriorated buildings (low improvement value)
- Relatively intact land values (land cannot be disinvested)
- High **land-to-total-value ratios**

Why Majority-Black Neighborhoods Lose

The structural mechanism:

Decades of redlining and disinvestment have left majority-Black neighborhoods with:

- Deteriorated buildings (low improvement value)
- Relatively intact land values (land cannot be disinvested)
- High **land-to-total-value ratios**

LVT shifts the tax base from total value to land value only. High land-to-total ratios $\Rightarrow$ the LVT multiplier ($4.40\times$ in Cook County) is applied to a base that already represents most of the parcel's value.

Why Majority-Black Neighborhoods Lose

The structural mechanism:

Decades of redlining and disinvestment have left majority-Black neighborhoods with:

- Deteriorated buildings (low improvement value)
- Relatively intact land values (land cannot be disinvested)
- High **land-to-total-value ratios**

LVT shifts the tax base from total value to land value only. High land-to-total ratios $\Rightarrow$ the LVT multiplier (4.40$\times$ in Cook County) is applied to a base that already represents most of the parcel's value.

Cook County GMM Cluster 4 (74 tracts):

- 75% Black, median income $41K
- Median land ratio: 0.45
- **96% median tax increase**; only 11% benefit
- South and West Side Chicago, formerly redlined

Why Majority-Black Neighborhoods Lose

The structural mechanism:

Decades of redlining and disinvestment have left majority-Black neighborhoods with:

- Deteriorated buildings (low improvement value)
- Relatively intact land values (land cannot be disinvested)
- High **land-to-total-value ratios**

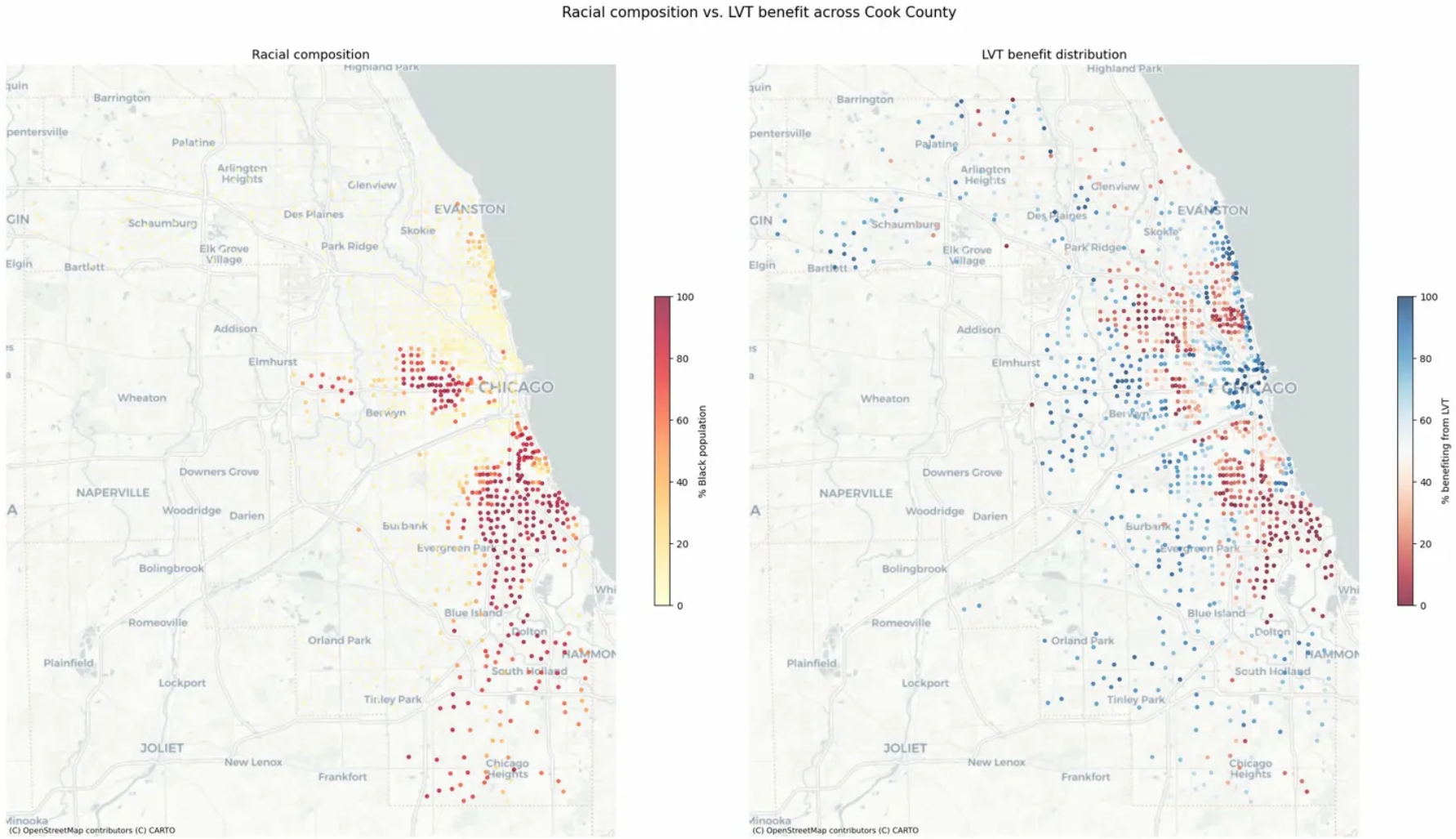
LVT shifts the tax base from total value to land value only. High land-to-total ratios $\Rightarrow$ the LVT multiplier (4.40$\times$ in Cook County) is applied to a base that already represents most of the parcel's value.

Cook County GMM Cluster 4 (74 tracts):

- 75% Black, median income $41K
- Median land ratio: 0.45
- **96% median tax increase**; only 11% benefit
- South and West Side Chicago, formerly redlined

The short-run incidence problem is a direct consequence of the long-run structural problem LVT is designed to solve.

Geographic Evidence: Race and LVT Burden Overlap



Left: racial composition by tract. Right: LVT benefit distribution. The geographic overlap between Black population concentration and LVT burden is visible on the South and West Sides, formerly redlined areas.

Who Owns the Losing Parcels?

Using 50.7M mailing address records from the CCAO, we identified owner type for parcels that **lose** under LVT in majority-Black neighborhoods:

	LVT Losers	LVT Winners
Absentee-owned	68.2%	62.5%
Institutional (10+ props)	12.9%	11.1%
Median props per owner	1	1

Who Owns the Losing Parcels?

Using 50.7M mailing address records from the CCAO, we identified owner type for parcels that **lose** under LVT in majority-Black neighborhoods:

	LVT Losers	LVT Winners
Absentee-owned	68.2%	62.5%
Institutional (10+ props)	12.9%	11.1%
Median props per owner	1	1

- The gap between losers and winners is only **5.7 percentage points**
- The burden is **not concentrated among large-scale speculators**
- It falls on small landlords, estates, and owner-occupants whose properties have high land ratios from decades of disinvestment
- This **reinforces the necessity of the homestead exemption**: the burden cannot be dismissed as falling on extractive actors alone

LVT Benefit Classification Results

City	Model	F1	AUC
Cook County	Logistic Regression	0.795	0.857
	SVM (Linear)	0.789	0.851
	Neural Network	**0.894**	**0.959**
New York City	Logistic Regression	0.381	0.765
	SVM (Linear)	0.382	0.750
	Neural Network	**0.678**	**0.875**
Philadelphia	Logistic Regression	0.752	0.816
	SVM (Linear)	0.749	0.813
	Neural Network	**0.787**	**0.870**

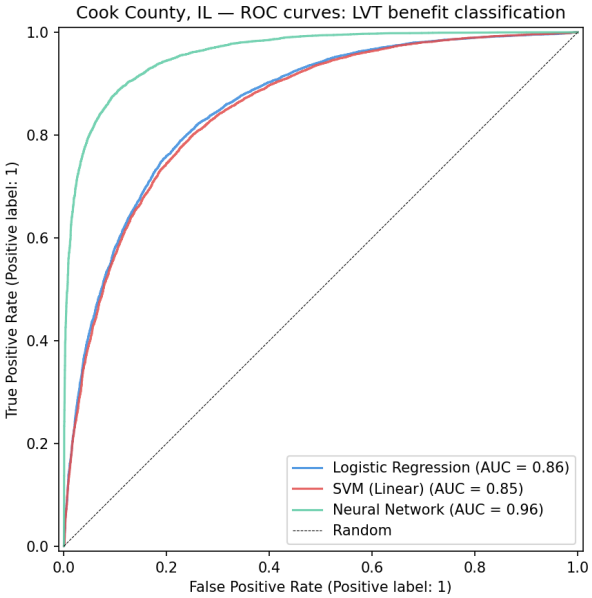
LVT Benefit Classification Results

City	Model	F1	AUC
Cook County	Logistic Regression	0.795	0.857
	SVM (Linear)	0.789	0.851
	Neural Network	**0.894**	**0.959**
New York City	Logistic Regression	0.381	0.765
	SVM (Linear)	0.382	0.750
	Neural Network	**0.678**	**0.875**
Philadelphia	Logistic Regression	0.752	0.816
	SVM (Linear)	0.749	0.813
	Neural Network	**0.787**	**0.870**

- **Neural network consistently outperforms** linear classifiers by **5–11 AUC points** across all three cities
- Consistent gap across three independent datasets ⇒ **genuine nonlinearity**: LVT benefit depends on interactions between building age, lot size, and neighborhood income
- Demographic variables carry **meaningful permutation importance** even after excluding land value components

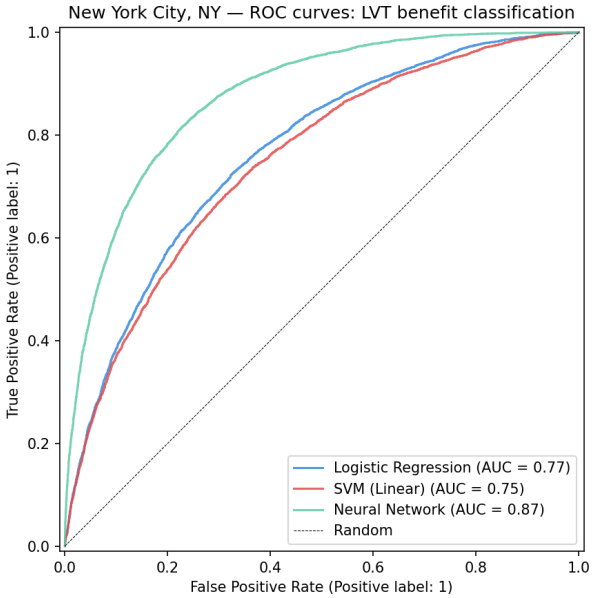
ROC Curves: Neural Network Advantage Across Cities

Cook County



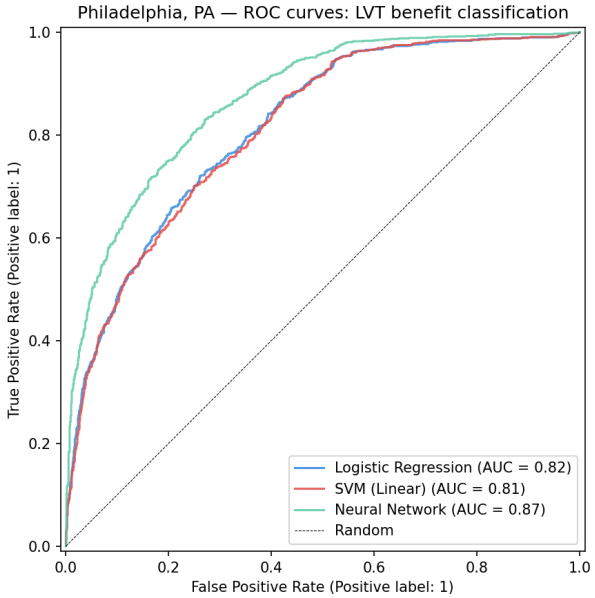
NN AUC = **0.959**

New York City



NN AUC = **0.875**

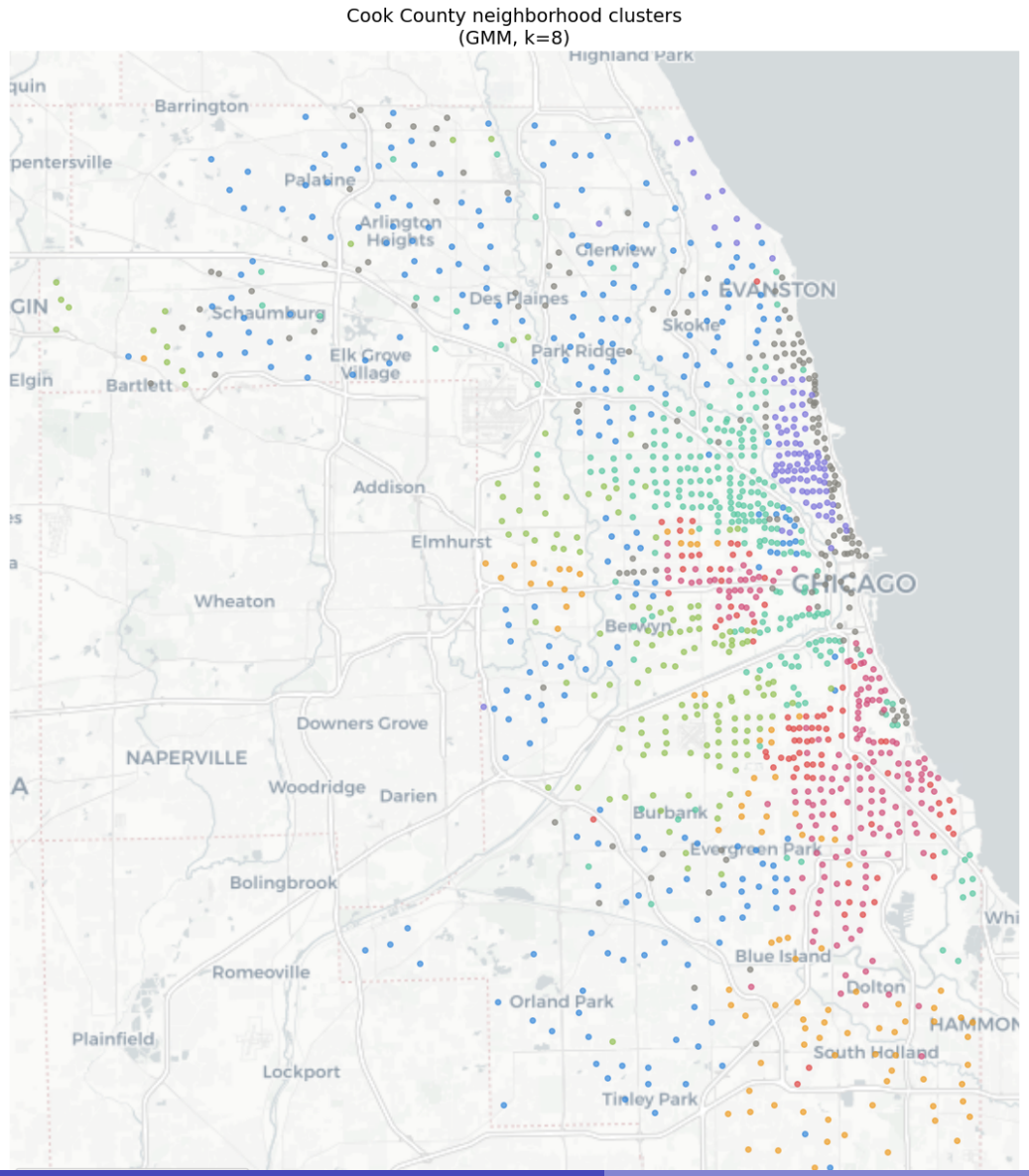
Philadelphia



NN AUC = **0.870**

The consistent NN advantage across three independent datasets with different feature sets confirms genuine nonlinearity in the relationship between observable property characteristics and LVT outcomes.

GMM Clustering: Unsupervised Recovery of Racial Geography



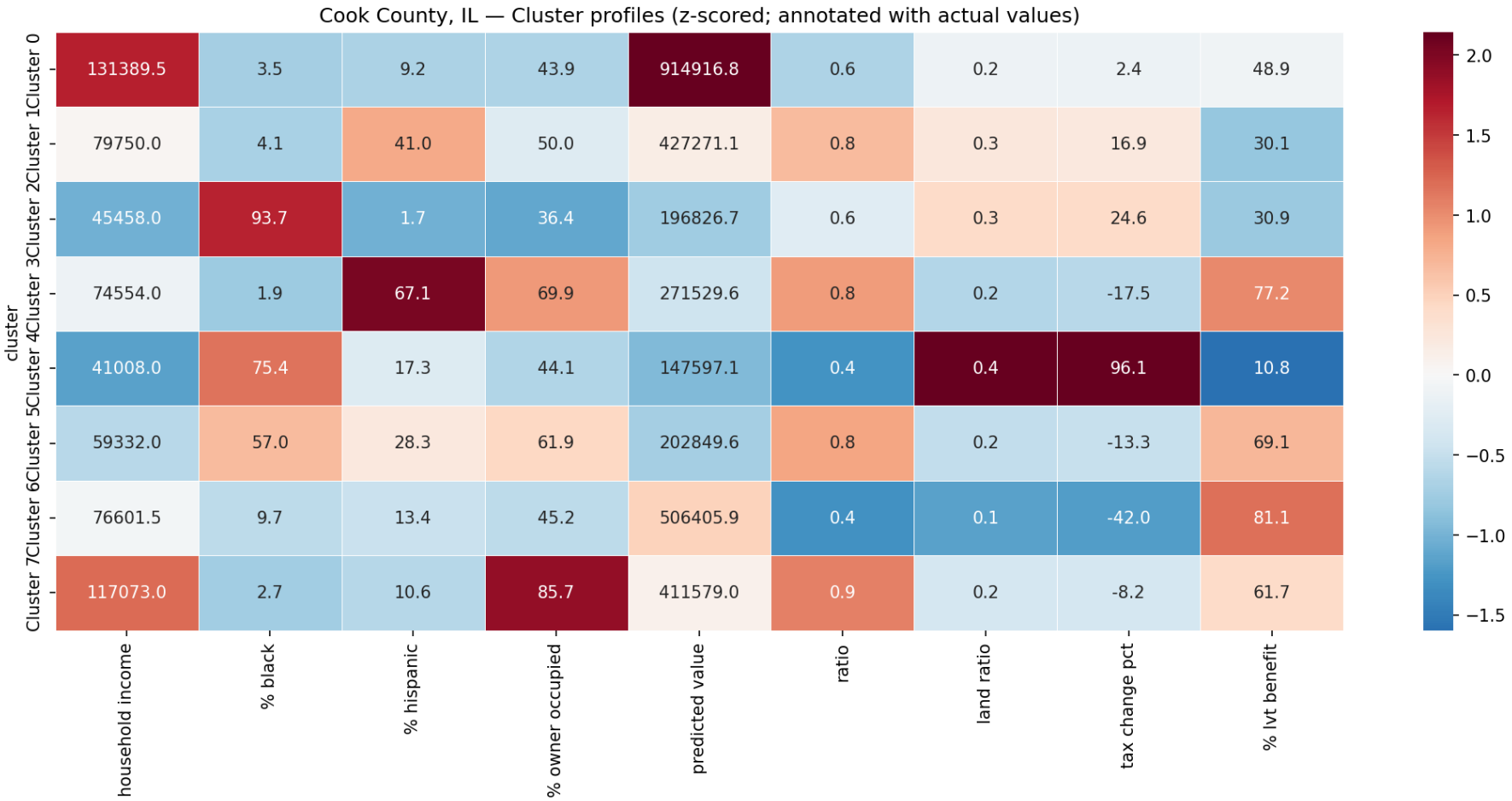
At $k = 2$:

Cluster A:
940 tracts, 3.8% Black
$88K income, 62.5% benefit

Cluster B:
330 tracts, 85.8% Black
$47K income, 35.0% benefit

No racial data was provided as input. The GMM recovers racial segregation from property characteristics alone, indicating that decades of segregation have produced property structures that encode racial geography.

Cluster Profiles (k=8, BIC-Optimal)



At $k = 8$, majority-Black neighborhoods split: middle-class Black tracts benefit from LVT (**76%**), while deeply segregated low-income Black tracts are hurt (**11–31%**). This distinction is invisible at $k = 2$ but essential for policy design.

Discussion: Three City-Specific Mechanisms

- **Cook County:** Assessment disparities operate through *spatial disadvantage* embedded in assessor models (transit access, flood risk, foreclosure density are themselves products of segregation). PRB is within acceptable range; equity gap is real but mechanism is place-based, not assessor-driven.

Discussion: Three City-Specific Mechanisms

- **Cook County:** Assessment disparities operate through *spatial disadvantage* embedded in assessor models (transit access, flood risk, foreclosure density are themselves products of segregation). PRB is within acceptable range; equity gap is real but mechanism is place-based, not assessor-driven.
- **New York City:** NYC's statutory Class 1 assessment cap ($\leq 6\%$/year increase) creates the disparity *mechanically*: rapidly appreciating minority neighborhoods are constrained below the market-value target, independently of assessor behavior. Documented by the NYC Advisory Commission [7] and confirmed here at the parcel level.

Discussion: Three City-Specific Mechanisms

- **Cook County:** Assessment disparities operate through *spatial disadvantage* embedded in assessor models (transit access, flood risk, foreclosure density are themselves products of segregation). PRB is within acceptable range; equity gap is real but mechanism is place-based, not assessor-driven.
- **New York City:** NYC's statutory Class 1 assessment cap ($\leq 6\%$/year increase) creates the disparity *mechanically*: rapidly appreciating minority neighborhoods are constrained below the market-value target, independently of assessor behavior. Documented by the NYC Advisory Commission [7] and confirmed here at the parcel level.
- **Philadelphia:** OPA block-level land schedules produce near-zero cross-sectional land value variation. The LVT simulation reflects abatement mechanics (Act 76), not appraised land values. Gini improvement ($0.279 \rightarrow 0.186$) is real but narrower in scope than a true LVT would produce.

The Homestead Exemption Solution

Exemption	Maj. Black	Maj. Hispanic	Maj. White
$0 (pure LVT)	46.6%	57.8%	64.2%
$10,000	**98.2%**	**95.7%**	**83.8%**
$20,000	99.7%	99.8%	94.1%
$50,000	100.0%	100.0%	99.3%

The Homestead Exemption Solution

Exemption	Maj. Black	Maj. Hispanic	Maj. White
$0 (pure LVT)	46.6%	57.8%	64.2%
$10,000	**98.2%**	**95.7%**	**83.8%**
$20,000	99.7%	99.8%	94.1%
$50,000	100.0%	100.0%	99.3%

- A **$10,000 exemption** on land assessed value lifts majority-Black benefit from 46.6% to **98.2%**
- Revenue-neutral at each exemption level (rate recomputed on reduced taxable base)
- Works because land assessed values in redlined neighborhoods are modest ($6–8K at 10% residential level), so the exemption shields most Black homeowners entirely
- Functions as a nonlinear correction: caps exposure for low-land-value parcels while preserving LVT incentives on high-value land

Key Findings & Policy Implications

Findings:

1. All three cities **fail IAAO standards** for assessment uniformity (COD, PRD). NYC and Philadelphia show statistically significant PRB regressivity ($p < 0.001$).
2. A revenue-neutral LVT benefits the majority of parcels in **two of three cities** (Cook 61.3%, Philadelphia 58.2%). NYC (30.3%) is the exception due to extreme land scarcity.
3. **In no city do majority-Black neighborhoods benefit at or above the citywide rate.** Black-White gaps are 11.9–17.6 pp across cities.
4. Neural networks outperform linear classifiers by 5–11 AUC points, demonstrating genuine nonlinearity in the LVT benefit prediction task.

Key Findings & Policy Implications

Findings:

1. All three cities **fail IAAO standards** for assessment uniformity (COD, PRD). NYC and Philadelphia show statistically significant PRB regressivity ($p < 0.001$).
2. A revenue-neutral LVT benefits the majority of parcels in **two of three cities** (Cook 61.3%, Philadelphia 58.2%). NYC (30.3%) is the exception due to extreme land scarcity.
3. **In no city do majority-Black neighborhoods benefit at or above the citywide rate.** Black-White gaps are 11.9–17.6 pp across cities.
4. Neural networks outperform linear classifiers by 5–11 AUC points, demonstrating genuine nonlinearity in the LVT benefit prediction task.

Policy implication: Any property tax reform, however race-neutral in design, will produce racially uneven outcomes unless it explicitly accounts for the property structures that discrimination created. A $10,000 homestead exemption in Cook County lifts the majority-Black benefit rate from **46.6%** to **98.2%** while remaining revenue-neutral.

Limitations & Future Work

Key limitations:

- **Static analysis**: no behavioral responses modeled (construction activity, rent pass-through, land speculation changes)
- **NYC dollar figures** understated ∼50% for Class 1 due to placeholder composite rate; distributional *direction* is unaffected
- **Philadelphia** 64% census tract match rate limits power for equity OLS (245 tracts); LVT result driven by abatement mechanics, not appraised land values
- **Cook County** uses county-wide multiplier; per-district simulation would produce different patterns in extreme-rate townships
- **Renters** not modeled – cannot directly benefit from LVT tax decreases; pass-through effects unknown

Limitations & Future Work

Key limitations:

- **Static analysis**: no behavioral responses modeled (construction activity, rent pass-through, land speculation changes)
- **NYC dollar figures** understated ∼50% for Class 1 due to placeholder composite rate; distributional *direction* is unaffected
- **Philadelphia** 64% census tract match rate limits power for equity OLS (245 tracts); LVT result driven by abatement mechanics, not appraised land values
- **Cook County** uses county-wide multiplier; per-district simulation would produce different patterns in extreme-rate townships
- **Renters** not modeled – cannot directly benefit from LVT tax decreases; pass-through effects unknown

Future work:

- Extend pipeline to additional cities
- Model behavioral incidence effects (construction, relocation)
- Evaluate complementary policies: homestead exemptions, targeted abatements during transition
- Longitudinal assessment cycle analysis

References

- IAAO, *Standard on Ratio Studies*, Kansas City, MO, 2013.
- P. Ganong and D. Shoag, "Why Has Regional Income Convergence in the U.S. Declined?" *J. Urban Econ.*, **102**, 76–90, 2017.
- C. Avenancio-León and T. Howard, "The Assessment Gap: Racial Inequalities in Property Taxation," *Q. J. Econ.*, **137**(3), 1383–1434, 2022.
- F. Plassmann and T. N. Tideman, "A Markov Chain Monte Carlo Analysis of Two-Rate Property Taxes," *J. Urban Econ.*, **47**(2), 216–247, 2000.
- R. F. Dye and R. W. England, *Assessing the Theory and Practice of Land Value Taxation*, Lincoln Institute, 2010.
- Cook County Assessor's Office, *CCAO Modeling Repository*, GitHub, 2021. `https://github.com/ccao-data`
- NYC Advisory Commission on Property Tax Reform, *Preliminary Report*, January 2020.

Thank you.

Sponsor: James Frederiksen, Duke University